MIL-F-8975B
29 September 1982
Superseding
MIL-F-8975A
15 May 1978

MILITARY SPECIFICATION

FASTENERS, BLIND, HIGH STRENGTH, INSTALLATION FORMED, CORROSION RESISTANT STEEL, HEAT RESISTANT STEEL AND TITANIUM, GENERAL SPECIFICATION FOR

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers installation formed high strength, corrosion resistant, heat resistant steel and titanium fasteners which can be installed from only one side of an assembly.

1.2 Classification. Fasteners furnished under this specification shall be of the following types and classes:

Type I - Pull Type - Positive Mechanically Locked (see 6.4)
Type II - Threaded - Self Locking (see 6.4)

Class 1 - 95 KSI Fsu, Corrosion and Heat Resistant Steel
Class 2 - 75 KSI Fsu, Titanium

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications and standards. Unless otherwise specified, the following specifications and standards of the issue listed in that issue of the Department of Defense Index of Specifications and Standards (DoDISS) specified in the solicitation, form a part of this specification to the extent specified herein.

SPECIFICATIONS

FEDERAL

QQ-P-35 Passivation Treatments for Corrosion-Resisting Steel

QQ-P-416 Plating, Cadmium (Electrodeposited)

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Engineering Specifications and Standards Department (Code 93), Naval Air Engineering Center, Lakehurst, NJ 08733, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

FSC 5320

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SPECIFICATIONS (Continued)

FEDERAL (Continued)

PPP-H-1581	Hardware (Fasteners and Related Items), Packaging and Packing for Shipment and Storage of

MILITARY

MIL-I-6866	Inspection, Penetrant Method of
MIL-L-8937	Lubricant, Solid Film, Heat Cured
MIL-T-9047	Titanium and Titanium Alloy Bars, Forgings and Forging Stock
MIL-L-46010	Lubricant, Solid Film, Heat Cured, Corrosion Inhibiting
MIL-L-81329	Lubricant, Solid Film, Extreme Environment
MIL-L-87132	Lubricant, Cetyl Alcohol, 1 Hexadecanol, Application to Fasteners

(See supplement 1 for list of applicable specification sheets.)

STANDARDS

MILITARY

MIL-STD-105	Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-129	Marking for Shipment and Storage
MIL-STD-889	Dissimilar Metals
MIL-STD-1312	Fasteners, Test Methods

(See supplement 1 for list of applicable military standards.)

(Copies of specifications, standards, drawings and publications required by manufacturers in connection with specific acquisition functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. The issues of the documents which are indicated as DoD adopted shall be the issue listed in the current DoDISS and supplement thereto, if applicable.

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American National Standards Institute

ANSI B46.1	Surface Texture (Surface Roughness, Waviness and Lay)

(Copies of American National Standards may be obtained from the American National Standards Institute, 1430 Broadway, New York, New York 10018.)

Society of Automotive Engineers, Inc.

AMS 5639	Bars, Forgings, Tubing and Rings - 19 Cr 10 Ni Sol. Heat Treated
AMS 5641	Bars and Forgings - 18.5 Cr 10 Ni, Free Machining; Swaging or Upsetting Sol. Heat Treated
AMS 5731 and 5732	Bars, Forgings, Tubing and Rings - 15 Cr 26 Ni 1.3 Mo
AMS 5737	Bars, Forgings and Tubing - 15 Cr 26 Ni 1.3 Mo 2.1 Ti

(Copies of Aerospace Material Specifications may be obtained from the Society of Automotive Engineers, Inc., 400 Commonwealth Drive, Warrendale, PA 15096.)

2.3 Order of precedence. In the event of a conflict between the text of the specification and the references cited herein, the text of this specification shall take precedence.

3. REQUIREMENTS

3.1 Specification sheets. The individual item requirements shall be as specified herein and in accordance with the applicable specification sheets. In the event of any conflict between requirements of this specification and the specification sheets, the latter shall govern.

3.2 Qualification. The fasteners furnished under this specification shall be products which are qualified for listing on the applicable Qualified Products List at the time set for opening of bids (see 4.3 and 6.3).

3.2.1 Retention of qualification. To maintain status on a Qualified Products List, certification shall be submitted at two year intervals to indicate continued compliance with the requirements of this specification (see 4.3.2).

3.2.2 Product change. Any change in materials or design shall require requalification of the product.

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3.3 Design and construction (Type I). The fastener shall be of a multiple piece construction and furnished as an integral assembly. It shall consist of a bolt body, spindle and locking collar. It shall have a means, other than friction, of mechanically locking the spindle to the bolt body.

3.3.1 Locking feature. The locking feature (locking collar) of the fastener shall be a secured component of the assembly. The fastener shall not disassemble during normal handling. Locking collars may be slotted. The slot width shall not be greater than 0.020 inch after assembly.

3.3.2 Dimensions. The fastener dimensions shall conform to the applicable specification sheet or standard.

3.3.3 Roundness and concentricity of heads. Fastener heads shall be round and concentric within the limits specified in table I.

3.3.4 Blind head diameters. The blind head diameters shall be as specified in table II.

3.3.5 Grip lengths. Grip lengths for all diameters shall be in 1/16-inch increments.

3.3.6 Materials (Type I). Fasteners furnished under this specification shall be in accordance with the applicable specification sheet or standard.

3.3.6.1 Dissimilar materials. The use of dissimilar materials shall be in accordance with MIL-STD-889.

3.3.7 Lubricant. Lubricant may be used on any components, provided the lubricant is not detrimental to the corrosion resisting capabilities of the fastener. The lubricant may be Cetyl alcohol in accordance with MIL-L-87132; Lauric Acid, WWM Carbo Wax; or, dry film lubricant in accordance with MIL-L-8937. The lubricant shall be identified on the qualified products list and any subsequent changes in lubricant are subject to approval, in writing, from the Naval Air Development Center, Code 60132 (see 6.3).

3.3.8 Protective treatment. When specified, the steel shall be cadmium plated in accordance with QQ-P-416, Type II, Class 2.

3.3.9 Passivation. All corrosion and heat resistant steel components shall be passivated in accordance with QQ-P-35.

3.4 Performance characteristics (Type I).

3.4.1 Installation. Installation shall be accomplished by the spindle being pulled into the sleeve, forming a blind head on the back side of the assembly, and removing the pulling portion of the spindle. A means of visual inspection for proper installation shall be provided (see 4.6.1).

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3.4.2 Tensile strength and tensile preload. The ultimate tensile strength and tensile preload of the installed fastener shall be as specified in table III (see 4.5.3).

3.4.3 Sheet take-up. The fastener shall have sufficient sheet take-up or pull-together to meet the requirements of table IV, when tested in steel plates having a total thickness equal to the maximum grip of the fastener (see 4.5.4).

3.4.4 Ultimate single shear strength. The ultimate single shear strength of the installed fastener shall be as specified in table III (see 4.6.2).

3.4.5 Tension-tension fatigue. The fastener shall withstand an average of not less than 60,000 cycles, but no single specimen shall fail in less than 40,000 cycles (see 4.6.3).

3.4.6 Spindle retention. The spindle of the installed fastener shall withstand a steadily applied axial push out minimum load as specified in table IV (see 4.6.4).

3.5 Discontinuities (Type I).

3.5.1 Seam and tool marks. Except as specified in 3.5.2 and 3.5.3, fasteners may possess seams and tool marks on any of the components of the fasteners. Seams and tool marks shall not be greater than the limits specified in table V.

3.5.2 Fastener head. The fastener head shall not contain more than three imperfections (openings, seams, inclusions, nicks, gouges). The depth of the imperfection shall not be greater than the following:

a. On the top of the head exceeding twice the depth limits shown in table V.

b. On the periphery of the head exceeding four times the depth limits shown in table V for protruding and flush head fasteners.

c. On the bearing surface of the head to depths exceeding the table V values. Fastener surface shall not be mutilated so that firm seating is prevented.

3.5.3 Head shank junction. Longitudinal seams not greater than the limits of table V are permitted in the head shank junction. Slight tool marks or undercuts of depth not greater than the limits specified in table V are permitted provided they fair into the shank with no sharp scratches, gouges or corners.

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3.6 Design and construction (Type II). The fastener shall be of a multiple piece construction and furnished as an integral assembly. It shall consist of a nut body, a corebolt, and a sleeve. A visible means shall be provided to lock the installed components together. The dimensions, concentricity, roundness and marking requirements shall be as specified on the applicable specification sheet or standard.

3.6.1 Component heads. The heads of all corebolts and nuts shall be forged.

3.6.2 Threads. The threads shall be left hand threads and the corebolt thread shall be formed by a single rolling process.

3.6.3 Materials (Type II). Materials shall be as specified on the applicable specification sheet or standard.

3.6.4 Penetrant inspection. The fasteners shall be fluorescent penetrant inspected (see 4.5.2).

3.7 Performance characteristics (Type II).

3.7.1 Installation. The fasteners shall be capable of being properly installed by driving equipment available at commercial and Government facilities. In addition, separate attachable nose adapters shall be supplied by the vendors listed in the appendix of this specification. The installed fastener shall meet the visual acceptance criteria of figure 2. The corebolt break off limits, the formed sleeve outside diameter, the backside protrusion and the seating torque shall be as specified in table XIII (see 4.7.1).

3.7.1.1 Corebolt protrusion. Corebolt protrusion shall be the method of visual inspection. The protrusion shall be as specified in table XIII.

3.7.2 Ultimate tensile strength and tensile preload. The tensile strength and tensile preload of the driven fastener shall be as specified in table XI (see 4.5.3).

3.7.3 Sheet take-up. The fastener shall conform to the sheet take-up requirements of table XII when tested in steel plates having a total thickness equal to the maximum grip of the fastener (see 4.5.4).

3.7.4 Ultimate double shear strength. The shear strength of the fastener shall be as specified in table XI (see 4.7.2).

3.7.5 Tension-tension fatigue. The fasteners shall withstand an average of not less than 30,000 cycles, but no single specimen shall fail in less than 15,000 cycles (see table XII and 4.7.3). The test shall be discontinued if the fastener does not fail after 100,000 cycles. No value above 100,000 cycles shall be used in calculating the average.

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3.7.6 Vibration. Fasteners shall withstand not less than 30,000 cycles without failure. The relative rotation of the nut and corebolt shall not be greater than 360°. The fastener shall not develop cracks or broken segments (see 4.7.4).

3.7.7 Locking torque. The assembled fasteners shall develop locking torque values equal or greater than the minimum required torque in table XII (see 4.7.5).

3.7.8 Metallurgical properties, Class 2 (Titanium).

3.7.8.1 Hydrogen content. The hydrogen content in the titanium components shall not be greater than .0125 percent (125 PPM) (see 4.7.6.1).

3.7.8.2 Widmanstaetten - Alpha case. Titanium components shall show no evidence of widmanstaetten or alpha case (see figure 8, 4.7.6.2 and 4.7.6.3).

3.8 Discontinuities (Type II).

3.8.1 Cracks. Fastener components shall be free of cracks in any direction or location (see 6.4.2).

3.8.2 Laps and seams. Fastener components may possess laps and seams not exceeding the depths specified in table X, except in prohibited locations (see figure 2).

3.8.3 Nuts. Discontinuities shall not be permitted in the head to shank fillet. Laps, seams, nicks and gouges are acceptable on the non-bearing surface of the head provided the depth is not greater than two times the value specified in table X.

3.8.4 Corebolts. Discontinuities shall not be permitted in the head to shank fillet. Thread laps which are not permissible and thread laps which are permissible within the limits of table X are shown in figure 2. Laps, seams, nicks and gouges are acceptable on the non-bearing surface of the head provided the depth is not greater than three times the value specified in table X.

3.8.5 Sleeves. There shall be no longitudinal discontinuities in the sleeves.

3.9 Workmanship. The fasteners shall be free of burrs, scale, excessive seams and other defects which would adversely affect their intended use.

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4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. Qualification inspection (see 4.3).

b. Quality conformance inspection (see 4.4).

4.3 Qualification inspection. Qualification inspection shall consist of the inspections in table VI for Type I and table XIV for Type II.

4.3.1 Sampling. Qualification test samples shall consist of fasteners of the diameter for which qualification is desired. The quantities shall be as indicated in table VI for Type I and table XIV for Type II. The fasteners shall have a grip length adequate to accommodate the test fixtures. The samples shall be identified with the following information and forwarded to the destination designated by the activity responsible for qualification (see 6.3).

Sample for Qualification Testing

FASTENERS, BLIND, HIGH STRENGTH, INSTALLATION FORMED,
CORROSION RESISTANT STEEL, HEAT RESISTANT STEEL AND TITANIUM

Name of manufacturer
Manufacturer's part number
Type for which qualification is desired
Submitted by (name) (date) for qualification testing in accordance with MIL-F-8975 under authorization (reference letter of authorization)

4.3.1.1 Qualification of the flush head configuration constitutes qualification of the equivalent protruding head configuration.

4.3.1.2 Certified test report. Certified test reports shall accompany the qualification samples. The reports shall include actual results of the tests specified herein. The report shall be accompanied by the applicable drawings.

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4.3.2 Retention of qualification. Certification shall be requested by NAVAIRDEVCEN (Naval Air Development Center), Code 6013, Warminster, PA 18974, from each manufacturer. NAVAIRDEVCEN will forward certification to NASC (Naval Air Systems Command) AIR-5303B. Certification shall be at the time of the two year review and shall be signed by a responsible official of management, attesting that the listed product(s) is still available from the listed plant, can be produced under the same conditions as originally qualified (i.e., same process, materials, construction, design, manufacturer's part number, or designation) and meets the requirements of the current issue of the specification. Failure to provide the certification will be cause for removal from the QPL. After completion of the certification review, the QPL shall be reprinted to show the date of validation. (DD Form 1718, Certification of Qualified Products, shall be used for obtaining certification.)

4.4 Quality conformance inspection. Quality conformance inspection shall be as specified in table VII for Type I and table XV for Type II.

4.4.1 Inspection lot. An assembly inspection lot shall consist of finished assembled components of the same configuration, diameter, and heat of material. The inspection lot assemblies and their components shall have been manufactured under the same conditions, heat treated in the same manner in one continuous run, and presented for delivery at the same time.

4.4.1.1 Plating lot. A plating lot shall consist of all parts processed in the plating bath within a 24 hour period. Cadmium plating shall be in accordance with QQ-P-416, Type II, Class 2.

4.4.2 Sampling. Samples shall be selected at random in accordance with MIL-STD-105, Inspection Level and Acceptance Quality Level (AQL) as specified in table VII for Type I and table XV for Type II. Identical sample items may be used for any of the tests, provided selection of random samples is maintained and known characteristics of the sample are not used to influence the integrity of the test results. Reduced sampling level may be instituted after five consecutive lots have passed the acceptance criteria.

4.4.3 Manufacturer's inspection report. Each inspection lot shall be accompanied by a copy of the manufacturer's inspection report, signed by an authorized representative of the manufacturer. The report shall be retained by the manufacturer (for five years) and made available, upon request, to the government. The report shall state that the fasteners are from an assembly lot which was manufactured and inspected in accordance with the requirements of this specification. The report shall identify the part number, assembly lot number and component lot numbers; and shall include the actual test results of each of the tests specified herein. In addition, the report shall contain information concerning the test procedures and instrumentation used in the quality conformance inspections.

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4.4.4 Material certification. Mill certification on all materials shall be kept on file and available to the Government.

4.5 Inspection methods (Type I and II).

4.5.1 Examination. Each of the sample fasteners, selected at random in accordance with tables VI and VII for Type I and tables XIV and XV for Type II shall be examined for conformance to the requirements for dimensions, concentricity, roundness, locking feature integrity, protective treatment, and marking. These examinations shall be visual. Optical aids or special gages shall be used when necessary. All dimensions are considered defective when out of tolerance.

4.5.2 Penetrant inspection. Fasteners shall be fluorescent penetrant inspected in accordance with MIL-I-6866, Type I, Method B. Penetrant inspection shall be performed before plating or with plating removed. Fasteners shall not be marked for identification of penetrant inspection. If penetrant indications are apparent, representative samples showing the indications shall be sectioned, mounted and microscopically examined for conformance to 3.5 or 3.8 as applicable.

4.5.3 Ultimate tensile strength and tensile preload. Tensile strength tests and tensile preload tests shall be performed in accordance with MIL-STD-1312, Test 8 and 16 except load deflection curves are required for qualification tests only. Tensile strength and preload tests are not applicable to fasteners having a grip length of less than twice the nominal diameter.

4.5.4 Sheet take-up. Each fastener shall be installed in a fixture similar to figure 7. Hole diameters for fasteners shall be in accordance with table VIII for Type I and table XIII for Type II. The nut supporting the lower plate shall be adjusted to provide a gap between the plates as specified in table IV for Type I and table XII for Type II. Each fastener shall engage the lower plate of the test fixture and shall clamp it against the upper plate when driven.

4.6 Inspection methods (Type I).

4.6.1 Installation (Type I). Fasteners shall be driven in minimum grip +.000 -.002 and maximum grip +.002 -.000 in holes drilled within tolerances specified in table VIII. Installed fasteners shall be visually examined for conformance to figure 1 and table II. The installed fasteners shall be visually inspected to establish that discontinuities of the blind head do not exceed the following limitations:

a. Longitudinal discontinuities shall not be greater than three times the limits of table V.

b. Circumferential discontinuities shall not be greater than twice the limits of table V.

c. General folds shall not be greater than twice the limits of table V.

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4.6.2 Ultimate single shear strength (Type I). Ultimate single shear tests shall be performed within the specified grip range of the fasteners in accordance with MIL-STD-1312, Test 20. Shear tests are not applicable to fasteners having a grip less than one diameter for protruding head type and one and one-half diameters for flush head type. Fasteners subjected to this test shall meet minimum loads specified in table III. The test need not be continued to destruction.

4.6.3 Tension-tension fatigue (Type I). Tension-tension fatigue tests shall be conducted using a thimble type fixture in accordance with figure 6. Hole diameters shall be within the limits specified in table VIII. Fasteners shall be tested in minimum grip -.002 +.000 and in maximum grip -.000 +.002. Set-up in figure 6 allows for visual inspection of blind head after driving. Fasteners in this test shall have the preload eliminated by removal of a low temperature melting washer (Cerrolow or equivalent 1/32 inch thick) from between the thimble and shim. Fatigue testing machine and test set-up shall conform to MIL-STD-1312, Test 11.

Test loading shall cycle between the minimum and maximum loads shown in table IV. The cycling speed shall not be greater than 3,600 cycles per minute and test shall be performed at room temperature. Tests shall not be conducted beyond 200,000 cycles.

4.6.4 Spindle retention (Type I). The spindle retention test shall be performed using test plates and fixtures in accordance with MIL-STD-1312, Test 8, as modified in figure 9. The hole diameter shall be within limits specified in table VIII. The load shall be steadily applied, at the rate shown in table IX, to the top of the spindle from the top of the fastener. The lock shall withstand the minimum axial pushout load values specified in table IV. Caution shall be exercised to ensure that mandrel force is applied directly in line with the axis of the spindle. A means of accurately determining the force applied to the mandrel shall be provided. Fasteners having a grip length of less than one diameter shall be tested with a bushing support having a clearance of approximately 1/32-inch around the blind head.

4.7 Inspection methods (Type II).

4.7.1 Installation (Type II). The fasteners shall be driven in hardened steel plates having a hardness of not less than 46 HRC and a surface finish on the blind side of 63-125 R_a in accordance with ANSI B46.1. The fasteners shall be driven in the maximum grip +.002 -.000 and the minimum grip +.000 -.002. The installation hole sizes shall be in accordance with table XIII and the countersinks for flush head bolts shall be equal to the maximum theoretical sharp head diameter +.003. The installed fastener shall be examined for conformance to the requirements shown in figure 3 and table XIII. The seating torque shall be checked on the minimum grip samples only.

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4.7.2 Ultimate double shear strength (Type II). The double shear test shall be conducted as follows:

a. Assemble the fastener until the corebolt breakneck groove is protruding through the head of the nut.

b. Teat the fastener as specified in MIL-STD-1312, Test 13.

c. Shear tests are not applicable to protruding head fasteners having a grip length of less than two diameters.

d. Shear tests are not applicable to flush head fasteners having a grip length of less than two and one-half diameters.

4.7.3 Tension-tension fatigue (Type II). The fastener shall be fatigue tested in accordance with MIL-STD-1312, Test 11. The fixture hole sizes shall be equal to the maximum shank diameter of the fastener. The fixture assembly shall include a removable shim so that preload may be eliminated before testing. The test need not be continued past 100,000 cycles.

4.7.4 Vibration (Type II). The fasteners shall be vibration tested in accordance with MIL-STD-1312, Test 7, except fixture hole sizes shall be equal to the maximum shank diameter of the fastener +0.004 -0.000.

4.7.5 Locking torque (Type II). The minimum locking torque shall be determined on an assembled fastener which has been driven in air to simulate an installed condition. One of the threaded components, either the nut or the corebolt, shall be retained in a vise or similar fixture to prevent it from rotating. The remaining threaded portion of the assembly shall then be rotated in the direction of disassembly with a torque wrench. The locking torque shall be recorded as the minimum reading of the torque wrench during the final 90o of the first revolution. The test shall be run at a rate slow enough to prevent an excessive increase in temperature of the bolt. Axial forces applied to the torque wrench shall be no more than necessary to maintain engagement of the wrenching surfaces.

4.7.6 Metallurgical conditions, Type II, Class 2 (titanium). Two samples from each heat treat lot of titanium components shall be longitudinally sectioned, mounted and microscopically examined for conformance to 3.7.8.1 and 3.7.8.2.

4.7.6.1 Hydrogen content, Class 2 (titanium). Remove material from the head-to-shank fillet area (titanium only) and determine the hydrogen content using procedures and equipment that are capable of analyzing hydrogen to an accuracy of .0010 percent (10 PPM). Plating and lubricant shall be removed before testing.

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4.7.6.2 Widmanstaetten. Titanium components shall show no evidence of widmanstaetten when tested in accordance with 4.7.6.

4.7.6.3 Alpha case, class 2. Titanium components shall be examined for alpha case. Components having alpha case greater than the limitations shown on figure 8 shall be rejected.

4.8 Inspection of preparation for delivery (Type I and II). The sampling and inspection of the preservation-packaging, packing and container marking shall be in accordance with PPP-H-1581 and MIL-STD-129.

5. PACKAGING

5.1 Preservation. Preservation shall be level A or C as specified (see 6.2).

5.1.1 Level A. The fasteners shall be preserved and packaged in accordance with the applicable level A requirement of PPP-H-1581.

5.1.2 Level C. The fasteners shall be preserved and packaged in accordance with the supplier's standard practice.

5.2 Packing. Packing shall be level A, B, or C as specified (see 6.2).

5.2.1 Level A. The fasteners shall be packaged in accordance with the applicable level A requirements of PPP-H-1581.

5.2.2 Level B. The fasteners shall be packaged in accordance with the applicable level B requirements of PPP-H-1581.

5.2.3 Level C. The fasteners shall be packaged in a manner which will insure arrival at destination in satisfactory condition and be acceptable to the carrier at lowest rates. Containers and packing shall comply with Uniform Freight Classification rules, or with National Motor Freight Classification rules.

5.3 Marking. In addition to any special marking specified in the contract, marking shall be in accordance with MIL-STD-129.

6. NOTES

6.1 Intended use. Fasteners covered by this specification are intended for structural attachments.

6.2 Ordering data.

6.2.1 Acquisition requirements. Acquisition documents should specify the following:

a. Title, number and date of this specification.

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b. Type and class of fastener (see 1.2).

c. Part number. (See applicable specification sheet.)

d. Selection of applicable levels of preservation and packing (see 5.1 and 5.2).

e. Special marking (see 5.3).

6.3 Qualification. With respect to products requiring qualification, awards will be made only for products which are, at the time set for opening of bids, qualified for inclusion in the Qualified Products List, whether or not such products have actually been so listed by that date. The attention of contractors is called to these requirements, and manufacturers are urged to arrange to have the products that they propose to offer to the Federal Government tested for qualification in order that they may be eligible to be awarded contracts or purchase orders for the products covered by this specification. The activity responsible for the Qualified Products List is the Naval Air Systems Command, Navy Department, Washington, D.C. 20361. However, information pertaining to qualification of products may be obtained from the Naval Air Development Center, Code 60132, Warminster, Pennsylvania 18974.

6.4 Definitions.

6.4.1 Type I and Type II fastener. Both Type I and Type II fasteners have design elements which maintain the integrity of the installed fasteners. The positive mechanical locking, Type I fastener has a locking element that keys the sleeve and pin together and thus prevents their disassembly when the structural joint is subjected to vibration and varying loads. The self-locking Type II fastener has a locking element which provides breakaway and prevailing frictional forces which prevent disassembly of nut and corebolt when the structural joint is subjected to vibration and varying loads.

6.4.2 Crack. A crack is defined as a clean crystalline break passing through the grain boundary with or without the inclusion of foreign elements.

Custodians:
Navy - AS
Air Force - 11
Army - AV

Review:
DLA - IS
Air Force - 99
Army - AR

Preparing Activity:
Navy - AS
(DoD Project No. 5320-0421)

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TABLE I. Tolerances on roundness and concentricity of heads, Type I.

Nominal Diameter (inch)	Flush Head	Protruding Head
5/32	0.010	0.015
3/16	.010	.015
1/4	.010	.020
5/16	.010	.020
3/8	.010	.025

TABLE II. Visual inspection tabulation, Type I.

Nominal Diameter (inch)	Spindle Position "B" (max)	Blind Head Diameter (min)	Collar Position "A" max (See figure 1)
5/32	0.020	0.195	.017
3/16	.024	.238	.022
1/4	.030	.315	.029
5/16	.038	.373	.037
3/8	.046	.448	.045

TABLE III. Mechanical properties (shear, tensile and tensile preload), Type I, Class 1.

Nominal Diameter	Single Shear Strength (lbs min)	Tensile Strength (lbs min)	Tensile Preload (lbs min)
5/32	1,980	1,150	115
3/16	2,925	1,690	169
1/4	5,000	2,900	290
5/16	7,200	4,170	417
3/8	10,380	5,970	597

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TABLE IV. Mechanical properties (tension-tension fatigue, sheet take-up and spindle retention), Type I.

Nominal Diameter	Tension-Tension Fatigue Loads (lbs) Class I		Sheet Take-Up (inch min)	Spindle Retention (lbs min)
	high	low		
5/32	519	52	.012	575
3/16	760	76	.015	845
1/4	1300	130	.020	1,440
5/16	1875	188	.025	2,080
3/8	2690	269	.032	2,980

TABLE V. Limits for surface discontinuity and tool mark depths, Type I.

Bolt Diameter (inch)	Depth (inch)
5/16 and under	0.005
3/8	.006

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TABLE VI. Qualification tests, Type I.

Test	Reqm't Para.	Method Para.	Sample Size Grip Condition		Sample Size	Recom. Grip	Number of Defectives Allowed
			min	max			
Examination	3.3	4.5.1	-	-	10	2D	0
Installation	3.4.1	4.6.1	10	10	20	2D	0
Tensile Strength and Preload 1/	3.4.2	4.5.3	5	5	10	2D	0
Sheet Take-Up	3.4.3	4.5.4	-	3	3	2D	0
Shear Strength	3.4.4	4.6.2	-	-	3	2D	0
Tension-Tension Fatigue	3.4.5	4.6.3	3	3	6	2D	0
Spindle Retention	3.4.6	4.6.4	-	-	5	2D	0

1/ Tensile strength and tensile preload may be run on the same samples.

TABLE VII. Quality conformance tests, Type I.

Test	Reqm't Para.	Method Para.	Sample Size (MIL-STD-105)		Acceptance Limits
			Normal	Reduced	
Examination	3.3	4.5.1	S-3	S-2	4.0
Installation	3.4.1	4.6.1	S-3	S-2	4.0
Tensile Strength	3.4.2	4.5.3	S-2	S-1	2.5
Shear Strength	3.4.4	4.6.2	S-2	S-1	2.5
Spindle Retention	3.4.6	4.6.4	S-2	S-1	2.5

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TABLE VIII. Hole tolerances, Type I.

Fastener Diameter (inch)	Hole Size (inch)
5/32	0.164 - 0.167
3/16	.199 - .202
1/4	.260 - .263
5/16	.312 - .315
3/8	.374 - .377

TABLE IX. Loading rates, Type I (lbs/minute + 10%).

Nominal Diameter (inch)	Spindle Retention	Preload Rate
5/32	1,150	230
3/16	1,700	340
1/4	2,900	580
5/16	4,100	840
3/8	5,900	1,200

TABLE X. Depth limits of permitted laps, seams, and discontinuities, Type II.

Component	Nominal Diameter				
	5/32	3/16	1/4	5/16	3/8
Nut	.005	.005	.005	.005	.006
Bolt	.002	.002	.004	.005	.005
Sleeve	.002	.002	.003	.004	.005

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TABLE XI. Mechanical properties (shear, tensile & tensile preload), Type II.

Nominal Diameter	Double Shear Strength (lbs min)		Tensile Strength (lbs min)		Tensile Preload (lbs min)	
	Class 1	Class 2	Class 1	Class 2	Class 1	Class 2
5/32	3,960	3,160	1,150	900	230	180
3/16	5,850	4,600	1,690	1,400	338	280
1/4	10,000	7,900	2,900	2,100	580	420
5/16	14,400	11,360	4,170	3,600	834	720
3/8	20,760	16,440	5,970	5,600	1,194	1,120

TABLE XII. Mechanical properties (tension-tension fatigue, sheet take-up and locking torque), Type II.

Nominal Diameter	Tension-Tension Fatigue Loads (lbs)				Sheet Take-Up (inch min)	Locking Torque (inch lbs min)
	Class 1		Class 2			
	high	low	high	low		
5/32	519	52	405	41	.032	1.0
3/16	760	76	630	63	.032	1.5
1/4	1300	130	945	95	.032	2.5
5/16	1875	188	1620	95	.032	3.5
3/8	2690	269	2520	252	.032	4.0

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TABLE XIII. Installation dimensions, Type II.

Nominal Diameter	Installation Hole Size	Chamfer max	Chamfer min	J dia max	K max	R Rad	X Breakoff (a) Flush Head	X Breakoff (a) Protruding Head	Seating Torque in lbs (b) min
5/32	.168 .165	.025	.015	.244	.272	.040 .030	+ .020 - .068	+.088 -.000	4
3/16	.202 .199	.030	.020	.300	.315	.040 .030	+ .015 - .073	+.098 +.010	6
1/4	.263 .260	.030	.020	.384	.356	.040 .030	+ .010 - .078	+.135 +.047	10
5/16	.315 .312	.035	.025	.427	.420	.050 .040	+ .010 - .083	+.146 +.043	20
3/8	.378 .375	.035	.025	.516	.495	.050 .040	+ .010 - .093	+.152 +.049	30

a. "X Breakoff" values are measured from the skin surface for protruding head; (+) values are aboye the skin surface, (-) values are below the surface. "X" values for flush head are measured from top of head.

b. Seating torque is the torque required to turn the bolt in a counterclockwise direction when the bolt is installed in the minimum grip condition.

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TABLE XIV. Qualification tests, Type II.

Test	Reqm't Para.	Method Para.	Sample Size Grip Condition			Total	Recom. Grip	Number of Defectives Allowed
			min	nom	max			
Examination	3.3	4.5.1	-	-	-	all qual samples	-	0
Penetrant Inspection	3.6.4	4.5.2	-	-	-	all qual samples	-	0
Installation	3.7.1	4.7.1	10	-	10	20	-	0
Tensile Strength & Preload 2/	3.7.2	4.5.3	5	-	5	2/ 10	2D min	0
Sheet Take-up	3.7.3	4.5.4	-	-	3	3	2D min	0
Shear Strength	3.7.4	4.7.2	-	-	-	3	2D min	0
Tension-Tension Fatigue	3.7.5	4.7.3	-	5	3	3	2D min	0
Vibration	3.7.6	4.7.4	-	5	-	5	4D min	0
Locking Torque	3.7.7	4.7.5	-	-	-	5	-	0
Metallurgical 1/	3.7.8	4.7.6	-	-	-	1/ 2	-	0
Hydrogen Content 1/	3.7.8.1	4.7.6.1	-	-	-	1/ 1	-	0

1/ Class 2 only. If hydrogen content is greater than the requirements, retest five additional samples. If any one of the additional samples is greater than the requirements, reject the lot.

2/ Tensile and tensile preload may be run on same samples.

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TABLE XV. Quality conformance tests, Type II.

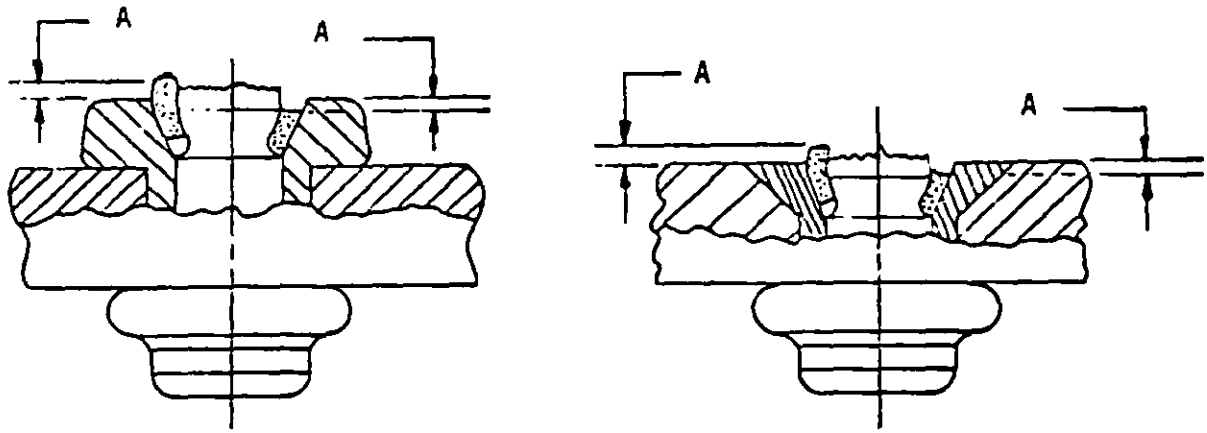
Test	Reqm't Para.	Method Para.	Sample Size 2/		Acceptance Limits
			Normal	Reduced	
Examination	3.3	4.5.1	S-3	S-2	2.5 AQL
Penetrant Inspection	3.6.4	4.5.2	S-3	S-2	1.5 AQL
Installation	3.7.1	4.7.1	S-3	S-2	2.5 AQL
Tensile Strength & Preload	3.7.2	4.5.3 (min grip)	S-2	S-1	0 Defectives
Shear Strength	3.7.4	4.7.2	S-2	S-1	0 Defectives
Locking Torque	3.7.7	4.7.5	10		0 Defectives
Hydrogen Content 1/	3.7.8.1	4.7.6.1	1		0 Defectives

1/ CLASS 2 ONLY. If hydrogen content is greater than the requirements, retest five additional samples. If one of the additional samples is greater than the requirements, reject the lot.

2/ Unless otherwise specified, sampling plan shall be in accordance with MIL-STD-105.

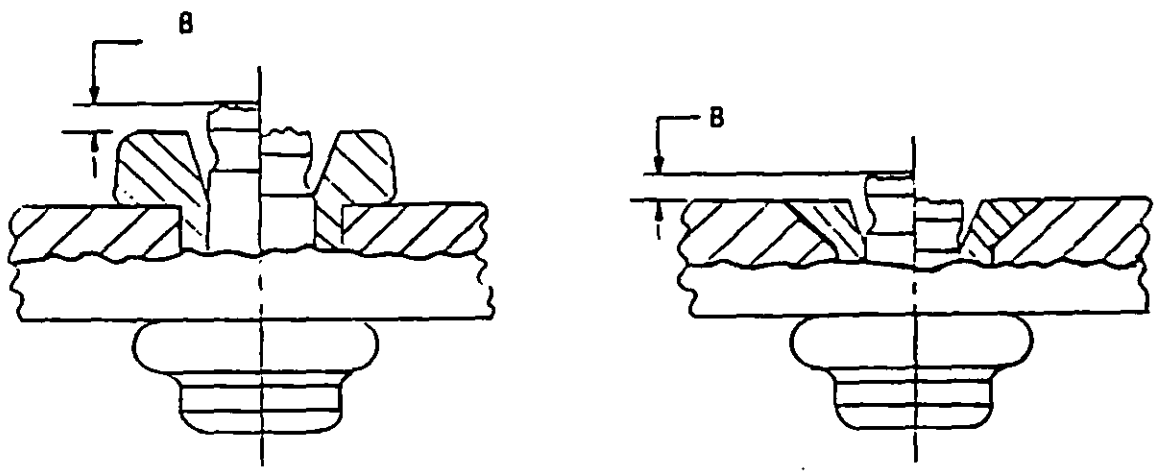
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INSPECTION OF LOCKING COLLAR POSITION



A . . . Maximum allowable distance of locking collar above or below fastener head.

INSPECTION OF SPINDLE POSITION



B . . . Maximum allowable distance above fastener head to nominal pin fracture surface. Nominal pin fracture surface may not recede below top of fastener head.

FIGURE 1. Visual inspection illustration, Type I.

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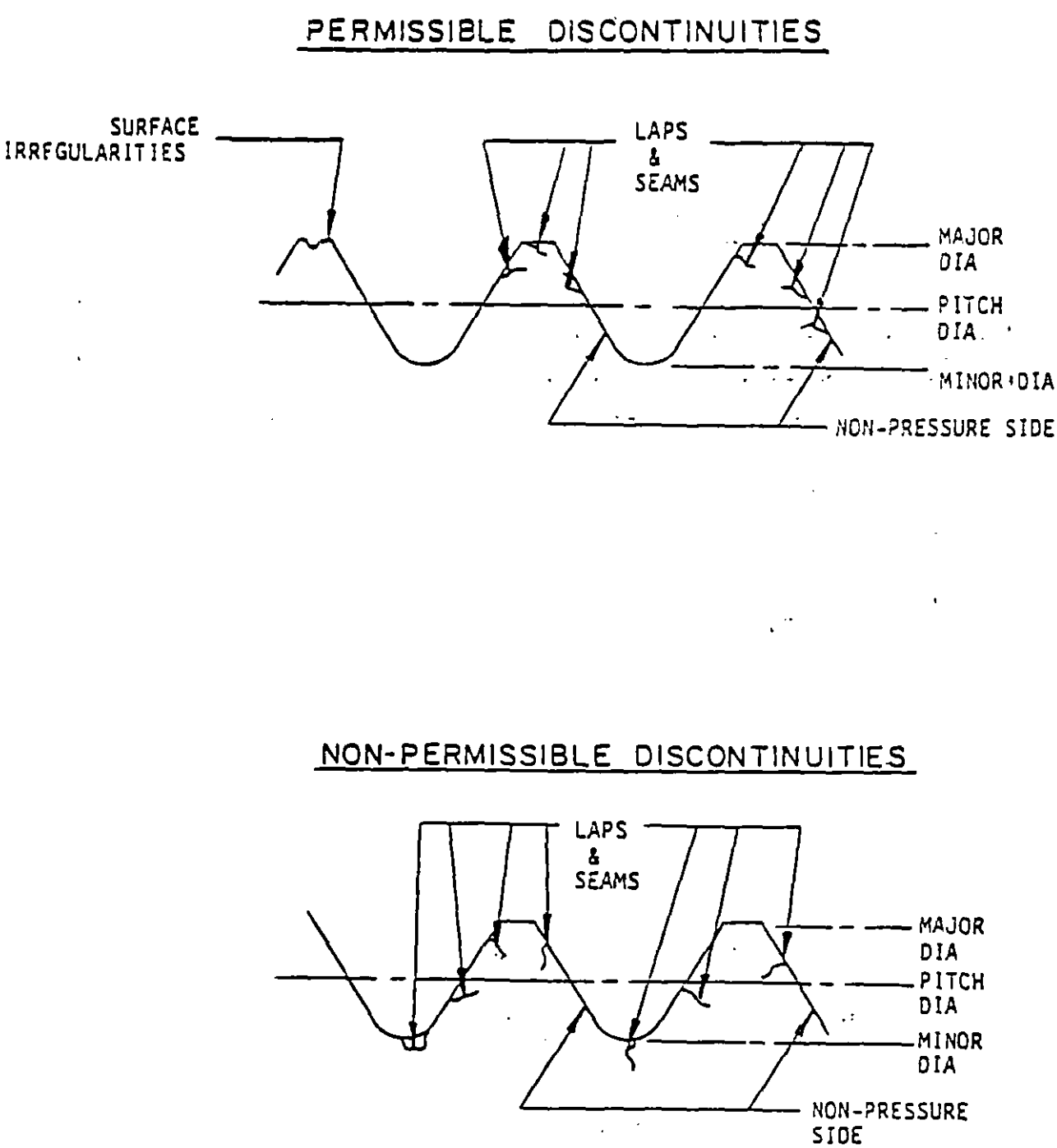


FIGURE 2. Fastener thread discontinuities, Type II.

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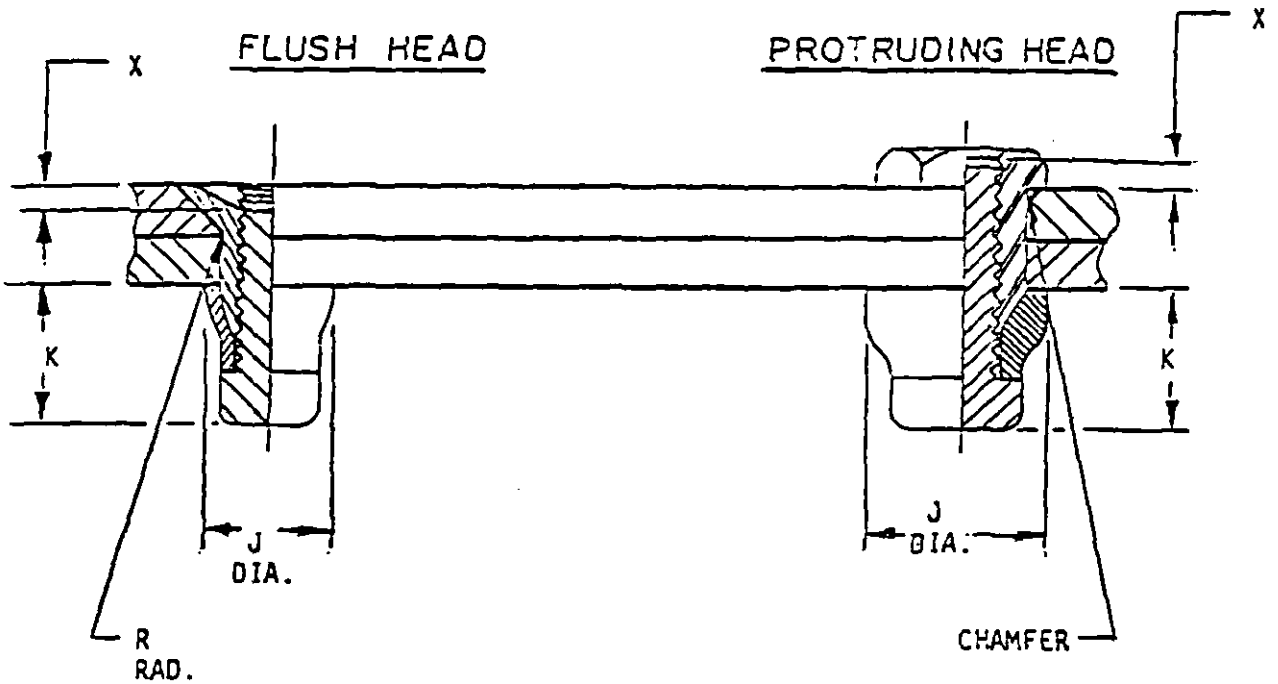
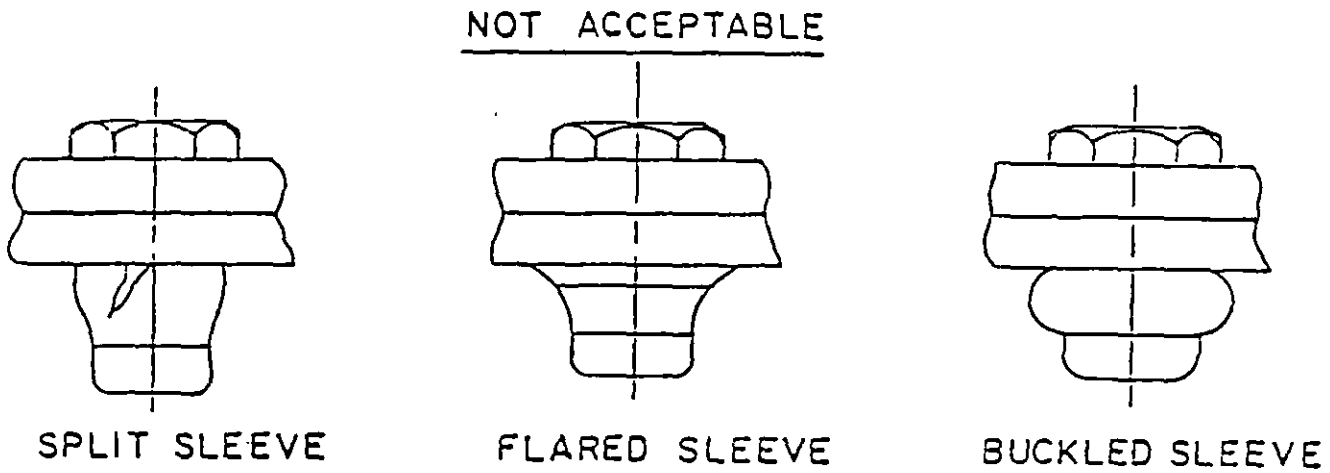


FIGURE 3. Installation dimension, Type II.

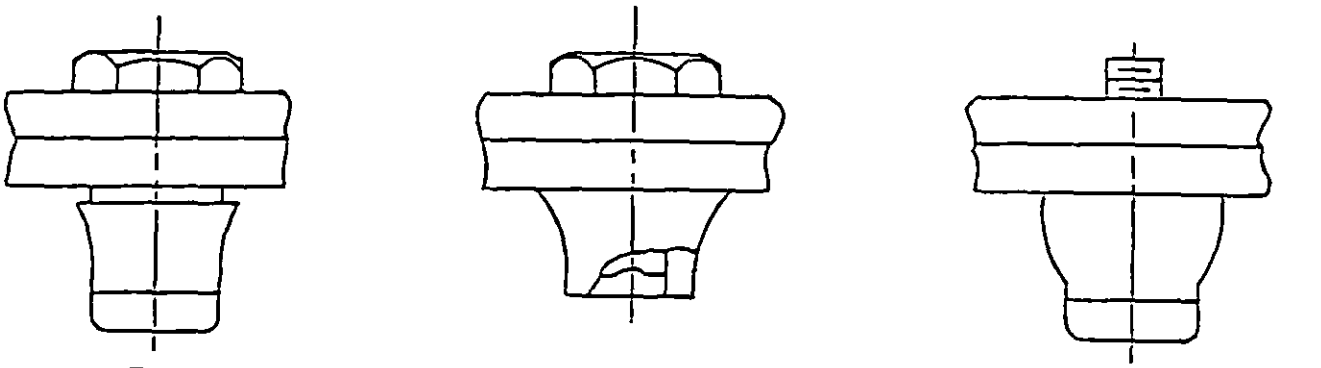
1. "X" Breakoff values are measured from the skin surface (for prot. head) (+) Values are above the skin surface (-) values are below the surface ("X" values are measured from top of head for flush head).

2. Seating torque is the torque required to turn the bolt in a counter-clockwise direction when the bolt is installed in the minimum grip condition.

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NOT ACCEPTABLE BEYOND
"J" DIA. AS SPECIFIED IN TABLE XIII



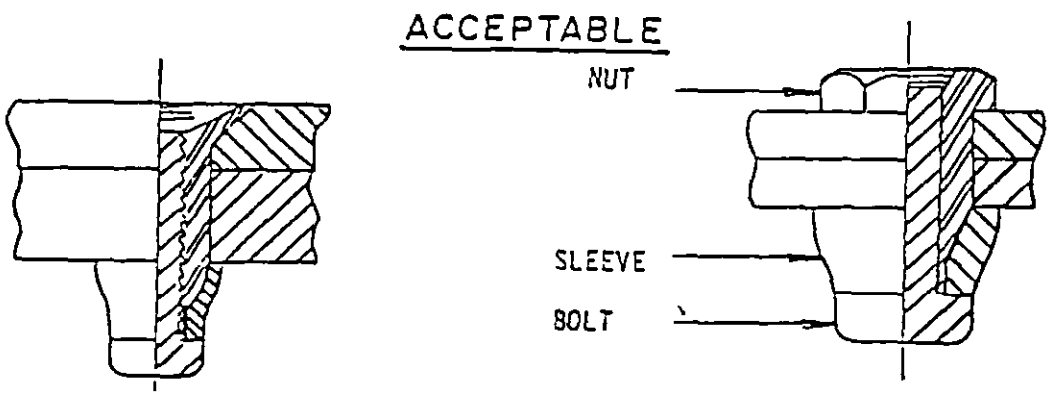


FIGURE 4. Installation - visual inspection, Type II.

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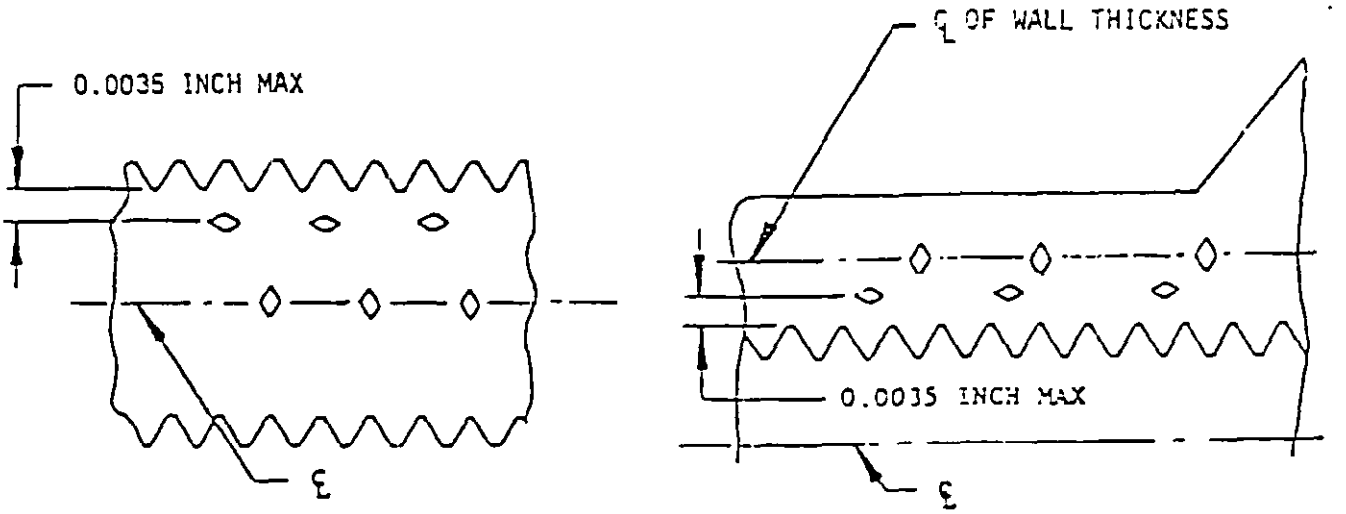


FIGURE 5. Location of microhardness impressions.

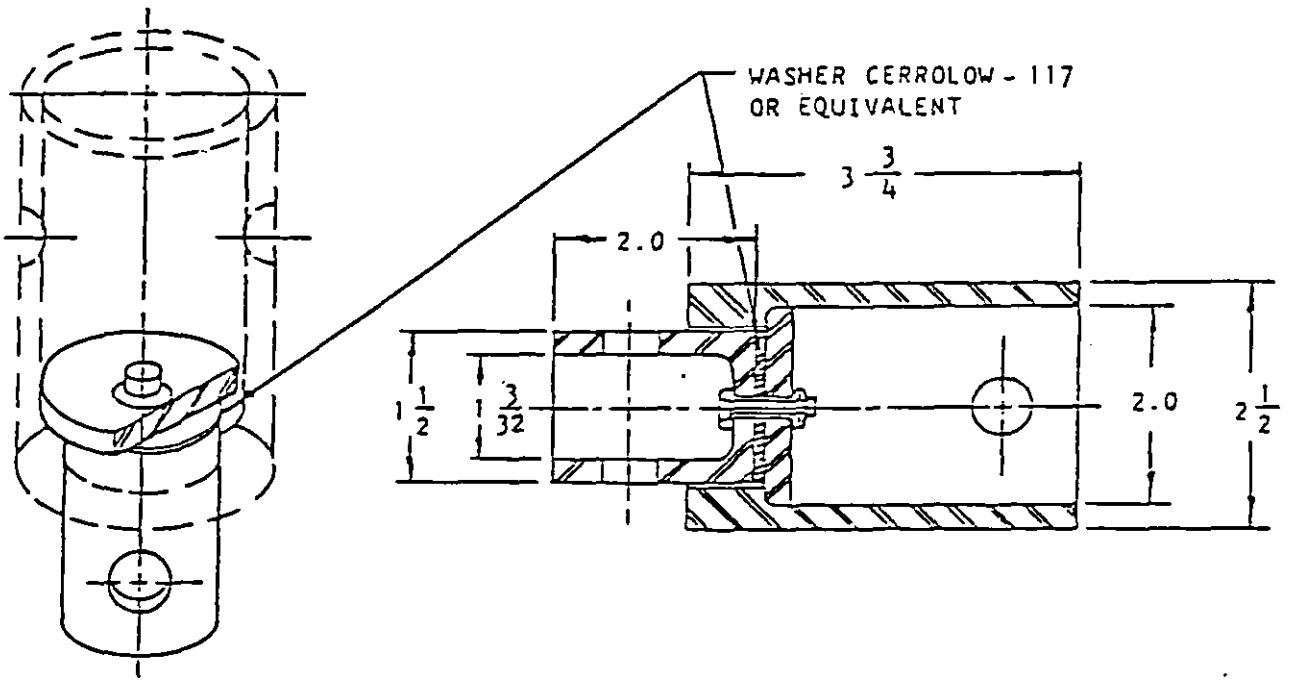


FIGURE 6. Tension-tension fatigue testing set-up, Type I.

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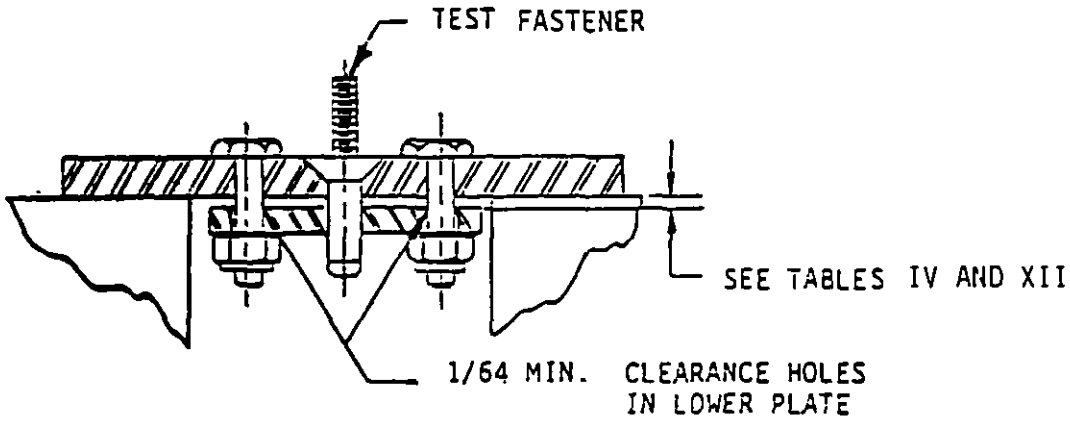
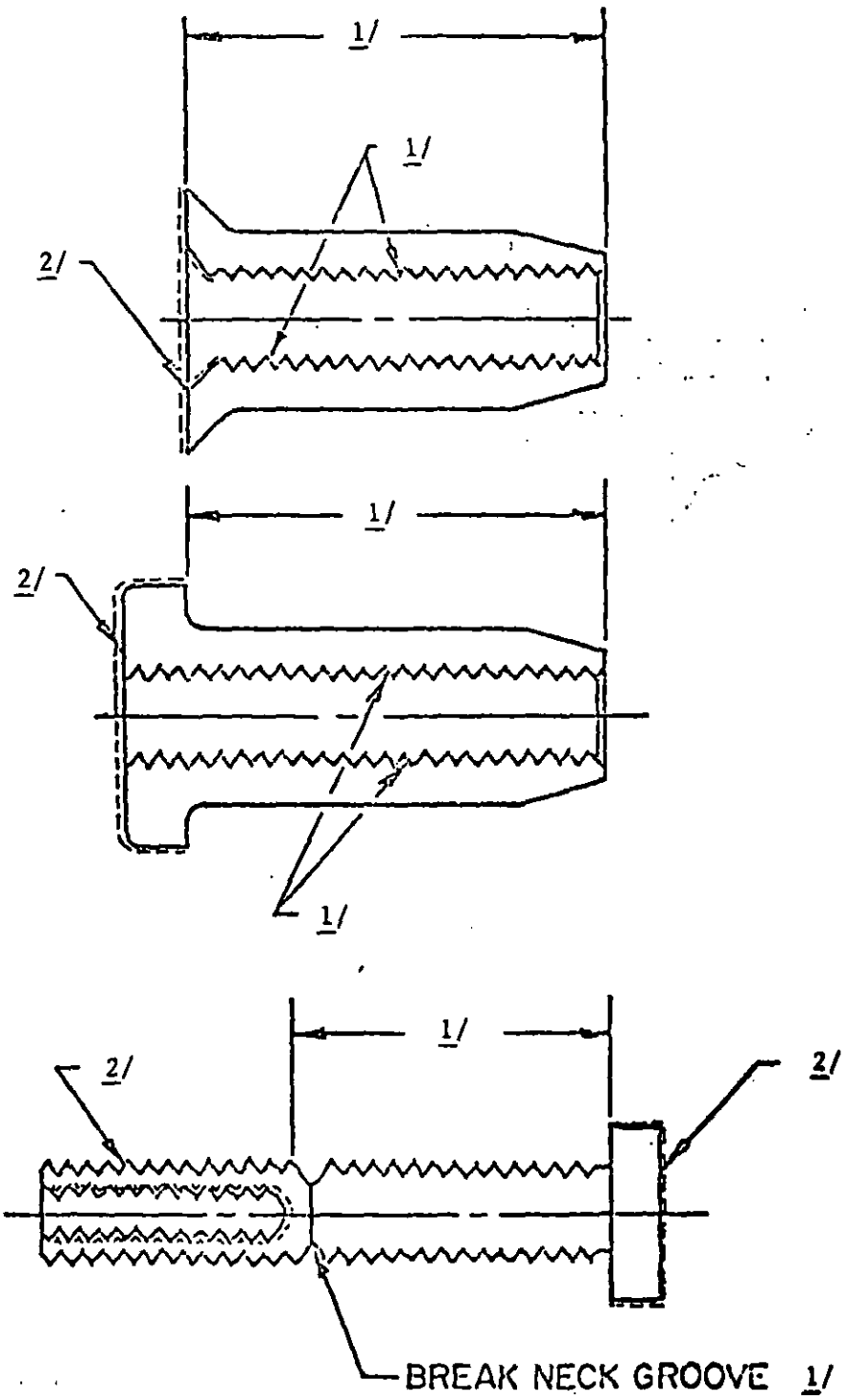


FIGURE 7. Sheet take-up fixture.

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NOTES:

1/ No alpha case permitted.

2/ Alpha case permitted to a max depth of .001.

FIGURE 8. Alpha case.

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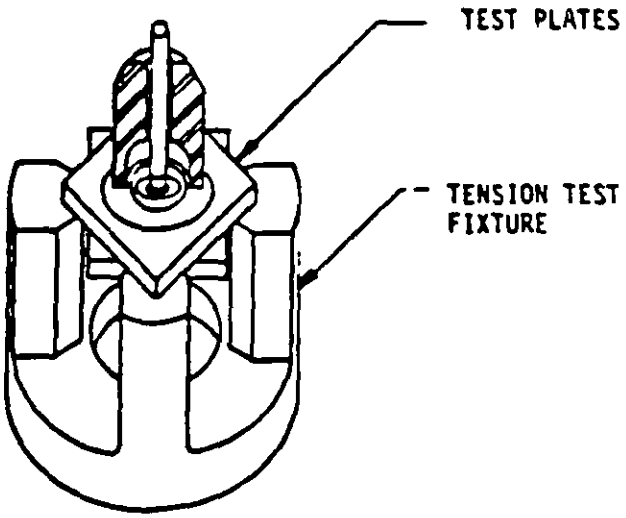


FIGURE 9. Spindle retention test set-up, Type I.

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APPENDIX A

INSTALLATION TOOLING FOR PULL TYPE FASTENERS (TYPE I)

Tooling Manufacturer: Huck Manufacturing Company, Federal Supply Code for Manufacturers (FSCM) 29666

10. SCOPE

10.1 Scope. This appendix covers tooling used for the installation of pull type fasteners.

MS21140 and MS21141	Nominal Size (Diameter)	Huck Manufacturing Co. Tool Part No. or equal	Nose Assembly Part Number or equal
-05	5/32	200	99-600
S05 or U05		223	99-680
S05 or U05		352	99-680
S05 or U05		206	99-904
S05 or U05	5/32	206	99-904-2
S05 or U05		609-1	99-853
S05 or U05		115	99-680
-06	3/16	200	99-589
S06 or U06		223	99-681
S06 or U06		352	99-681
S06 or U06		206	99-905
S06 or U06		206	99-905-2
S06 or U06		609-1	99-854
S06 or U06		115	99-681
-08 or U08		115	99-591
-08 or U08		207	99-906
-08 or U08		353	99-591
-08 or U08		354	99-591
-08 or U08		610-1	99-679
			99-838
			99-865
-08 or U08	1/4	611-1	99-682
-08 or U08		612-1	99-682
			99-838
			99-865
-08 or U08		4801	99-591
-08 or U08		5901	H99-662
-10 or U10		116	99-663
-10 or U10	5/16	4801	99-823
-10 or U10		5901	H99-663
-12 or U12		116	99-599
-12 or U12	3/8	4801	99-988
-12 or U12		5901	H99-599
-14 or U14	7/16	116	99-738
-14 or U14		5901	H99-738
-16 or U16	1/2	116	99-678
-16 or U16		5901	H99-678

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-05 and -06 signify Double Action (shifting) Tooling required for 5/32 and 3/16 only.

S05 and S06 signify Single Action (non-shifting) Tooling required for 5/32 and 3/16 only.

-08, -10, -12, -14 and -16 sizes have been and continue to be single action (non-shifting) installed.

U code fasteners utilize single action (non-shifting) tooling.

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APPENDIX B

INSTALLATION TOOLING FOR TYPE II BLIND FASTENER

Tooling Manufacturers:

1. Monogram Aerospace Fasteners, Federal Supply Code for Manufacturers (FSCM) 98524

2. Lok-Fast Incorporated, Federal Supply Code for Manufacturers (FSCM) 01022

10. SCOPE

10.1 Scope. This appendix covers tooling used in the installation of fasteners covered by this specification.

Tool Type		Pneumatic Driver, Pistol Grip			
		Monogram P/N MP550/Lok-Fast P/N 450 MTD			
		Driving Adapters			
		Monogram P/N		Lok-Fast P/N	
Fastener DIA	Head Type	Wrench Adapter	Nose Adapter	Wrench Adapter	Nose Adapter
5/32	Flush	MP-5	MPF-5	TD1651	PW3652
3/16	Flush	MP-6	MPF-6	TD2001	PW3002
1/4	Flush	MP-8	MPF-8	TD2601	PW3602
5/16	Flush	MP-10	MPF-10	TD3121	PW3122
3/8	Flush	MP-12	MPF-12	TD3751	PT3752
5/32	Protruding	MP-5	MPP-5	TD1651	PW3652P
3/16	Protruding	MP-6	MPP-6	TD2001	PW3002P
1/4	Protruding	MP-8	MPP-8	TD2601	PW3602P
5/16	Protruding	MP-10	MPP-10	TD3121	PW3122P
3/8	Protruding	MP-12	MPP-12	TD3751	PT3752P

NOTE: Complete tool consists of basic tool (pneumatic driver, pistol grip), one wrench adapter and one nose adapter.

EXAMPLE: MP550, MP-5 and MPF-5 components are required to install 5/32 Diameter Flush Head Blind Fasteners, Type II.

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Tool Type		Hand Tool			
		Monogram P/N MH75		Lok-Fast P/N HW200 HW300	
		Driving Adapters			
		Monogram P/N		Lok-Fast P/N	
Fastener DIA	Head Type	Wrench Adapter	Nose Adapter	Wrench Adapter	Nose Adapter
5/32	Flush	MH-5	MHF-5	HW1651	HW1652
3/16	Flush	MH-6	MHF-6	HW2001	HW2002
1/4	Flush	MH-8	MHF-8	HW2601	HW2602
5/16	Flush	MH-10	MHF-10	HW3121	HW3122
3/8	Flush	MH-12	MHF-12	HW3751	HW3752
5/32	Protruding	MH-5	MHP-5	HW1651	HW1652P
3/16	Protruding	MH-6	MHP-6	HW2001	HW2002P
1/4	Protruding	MH-8	MHP-8	HW2601	HW2602P
5/16	Protruding	MH-10	MHP-10	HW3121	HW3122P
3/8	Protruding	MH-12	MHP-12	HW3751	HW3752P

NOTE: Lok-Fast hand tool HW200 installs 5/32 DIA and 3/16 DIA blind fasteners; HW300 installs 1/4 DIA and 3/8 DIA blind fasteners.

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NAME OF ORGANIZATION AND ADDRESS OF SUBMITTER

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